

ControlSift: Reducing Inequality in Evidence Assurance with Small-Model Triage Research

Mentor Me Collective · Google Cloud Launchpad · DeepMind AI Research Foundations
Capstone

Official UN SDG: Goal 10 – Reduced Inequalities · Dataset v1.1.0 · Protocol protocol-v1-locked

Solo project · Primary metric: macro F1 · Seed 42

1. Problem statement and objectives

Organizations with limited GRC budgets face unequal access to high-quality evidence review. Template binders and policy PDFs are often treated as proof that controls operate. That false assurance harms smaller clinics, schools, nonprofits, and community providers more than enterprises that can buy deep audits.

Objective: Determine whether QLoRA adaptation of Gemma 3 1B Instruct improves five-class evidence-quality classification (SUFFICIENT, PARTIAL, INSUFFICIENT, IRRELEVANT, CONTRADICTORY) on a sealed synthetic benchmark—so triage methods can be published openly for under-resourced reviewers.

Scope: Research and education only. No production auditor. No private customer evidence. Human judgment remains authoritative.

2. Research and analysis

Problem identification used public cybersecurity and assurance sources (industry breach reporting, CISA/NIST guidance, public discourse on paper compliance). Modeling data is synthetic by design: 1,500 compositional evidence packets, family-isolated splits (1000/200/200/100), rule-derived labels with structural audit and

scripted spot-check. An early draft allowed TF-IDF to saturate; dataset v1.1 was hardened so lexical shortcuts no longer define the ceiling.

Classical results (sealed test): majority macro F1 pending; TF-IDF + logistic regression pending (challenge pending). Gemma zero-shot pending, few-shot pending, QLoRA pending. [Gemma TEST metrics stay null until after_kaggle refreshes docs/data/results.json — never invented.](#)

3. Solution development

Solution ladder: majority → TF-IDF → Gemma zero-shot → few-shot → Gemma 3 1B + QLoRA (4-bit NF4). Evaluation protocol frozen before LLM claims. Public research site includes Failure Lab, Assurance (Data Card, Model Card, Risk Register, Intended Use), and curriculum map against DeepMind AI Research Foundations.

Practicality: free-tier GPU packaging, smoke-then-full runs, open documentation. Sustainability: methods are inspectable and reproducible without enterprise spend—aligned with SDG 10 access goals.

4. Implementation plan

Phase 0 (complete): benchmark, seal, classical baselines, MMC hub. Phase 1–2: HF license + Kaggle smoke and prompted Gemma baselines. Phase 3: QLoRA train/eval. Phase 4: error analysis. Phase 5: finalize PDF, 8-slide deck, ≤5-minute video, form submission. Risks (access delay, OOM, overclaim, format miss) and mitigations are listed on the implementation-plan page.

5. Outcomes to date

Sealed protocol and public classical receipts.

Meaningful lexical ceiling after dataset harden (~0.53 macro F1).

Failure Lab + Assurance pack for honest communication.

Open items: Gemma ladder TEST metrics; final video recording for form upload.

6. Recommendations

Proceed with free-GPU execution; publish only from result files; keep human-in-the-loop boundaries; treat per-class weakness (especially SUFFICIENT) as first-class findings. Do not deploy as an audit autopilot.

7. Authorship attestation

This capstone is a human-directed applied research project. Assistive tools were used for drafting and engineering support; problem selection, experimental design, protocol seal, dataset decisions, and integrity rules are the author's responsibility. It is not an AI-generated project submission.